

## T3 group factors: EFT\_1\_after\_F

Active theory: SM +  $S_1 + S_2 + C_{LLS_1S_2}$ . The fermion  $F$  is absent, so  $y_1$ ,  $y_2$  and  $M_F$  are not running EFT couplings.

$$j = \frac{d-1}{2}, \quad C_2(d) = \frac{d^2-1}{4}, \quad T(d) = \frac{d(d^2-1)}{12}.$$

In every comparison table,  $\mathcal{S}$  denotes the RGE structure multiplying the displayed group factor in  $16\pi^2\beta$ .

## Notation key

| symbol                    | interaction / definition                                           |
|---------------------------|--------------------------------------------------------------------|
| $g_1, g_2, g_3$           | SM gauge interactions                                              |
| $Y_u, Y_d, Y_e$           | SM Yukawa interactions                                             |
| $y_1$                     | $LF S_1 + \text{h.c.}$                                             |
| $y_2$                     | $LF S_2 + \text{h.c.}$                                             |
| $M_F$                     | $-\bar{F} M_F F$                                                   |
| $T$                       | $\text{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$ |
| $T_\nu^{(1)}$             | $\text{Tr}(y_1^\dagger y_1)$                                       |
| $T_\nu^{(2)}$             | $\text{Tr}(y_2^\dagger y_2)$                                       |
| $T_\nu$                   | $T_\nu^{(1)} + T_\nu^{(2)}$                                        |
| $m_1^2, m_2^2$            | $m_1^2 S_1^\dagger S_1, m_2^2 S_2^\dagger S_2$                     |
| $\lambda_1$               | $\frac{1}{2} \lambda_1 (H^\dagger H)^2$                            |
| $\lambda_{S_i}^{(1)}$     | $\frac{1}{2} \lambda_{S_i}^{(1)} (S_i^\dagger S_i)^2$              |
| $\lambda_{HS_i}^{(1)}$    | $(H^\dagger H)(S_i^\dagger S_i)$                                   |
| $\lambda_{HS_i}^{(A)}$    | $(H^\dagger T^A H)(S_i^\dagger T^A S_i)$                           |
| $\lambda_5$               | $HH S_1 S_2^\dagger + \text{h.c.}$                                 |
| $\lambda_{12}^{(1)}$      | $(S_1^\dagger S_1)(S_2^\dagger S_2)$                               |
| $\lambda_{12}^{(A)}$      | $(S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$                       |
| $\lambda_{12}^{(\times)}$ | $(S_1^\dagger S_2)(S_2^\dagger S_1)$                               |

## Stage field content

| run  | $S_1$              | $S_2$              | $F$      | integrated at this stage | active after matching |
|------|--------------------|--------------------|----------|--------------------------|-----------------------|
| T3-B | $(2, \frac{1}{2})$ | $(2, \frac{3}{2})$ | $(1, 1)$ | F                        | S1, S2                |

## Field representations

| run    | $S_1$              | $S_2$              | $\alpha$ |
|--------|--------------------|--------------------|----------|
| T3 - B | $(2, \frac{1}{2})$ | $(2, \frac{3}{2})$ | 1        |

## Gauge and Yukawa beta functions in this EFT

$\beta_{g_1}^{(1)}$

| $\mathcal{S}$ | T3 - B ( $\alpha = 1$ ) |
|---------------|-------------------------|
| $g_1^3$       | $\frac{17}{2}$          |

$\beta_{g_2}^{(1)}$ 

| $\mathcal{S}$ | T3 – B ( $\alpha = 1$ ) |
|---------------|-------------------------|
| $g_2^3$       | $-\frac{17}{6}$         |

 $\beta_{g_3}^{(1)}$ 

| $\mathcal{S}$ | T3 – B ( $\alpha = 1$ ) |
|---------------|-------------------------|
| $g_3^3$       | $-7$                    |

 $\beta_{Y_u}^{(1)}$ 

| $\mathcal{S}$                 | T3 – B ( $\alpha = 1$ ) |
|-------------------------------|-------------------------|
| $(Y_d(Y_d)^\dagger Y_u)_{ij}$ | $-\frac{3}{2}$          |
| $(Y_u(Y_u)^\dagger Y_u)_{ij}$ | $\frac{3}{2}$           |
| $g_2^2(Y_u)_{ij}$             | $-\frac{9}{4}$          |
| $g_3^2(Y_u)_{ij}$             | $-8$                    |
| $g_1^2(Y_u)_{ij}$             | $-\frac{17}{12}$        |
| $T(Y_u)_{ij}$                 | $1$                     |

 $\beta_{Y_d}^{(1)}$ 

| $\mathcal{S}$                 | T3 – B ( $\alpha = 1$ ) |
|-------------------------------|-------------------------|
| $(Y_d(Y_d)^\dagger Y_d)_{ij}$ | $\frac{3}{2}$           |
| $(Y_u(Y_u)^\dagger Y_d)_{ij}$ | $-\frac{3}{2}$          |
| $g_2^2(Y_d)_{ij}$             | $-\frac{9}{4}$          |
| $g_3^2(Y_d)_{ij}$             | $-8$                    |
| $g_1^2(Y_d)_{ij}$             | $-\frac{5}{12}$         |
| $T(Y_d)_{ij}$                 | $1$                     |

 $\beta_{Y_e}^{(1)}$ 

| $\mathcal{S}$                 | T3 – B ( $\alpha = 1$ ) |
|-------------------------------|-------------------------|
| $(Y_e(Y_e)^\dagger Y_e)_{ij}$ | $\frac{3}{2}$           |
| $g_2^2(Y_e)_{ij}$             | $-\frac{9}{4}$          |
| $g_1^2(Y_e)_{ij}$             | $-\frac{15}{4}$         |
| $T(Y_e)_{ij}$                 | $1$                     |

 $\beta_{m_1^2}$ 

| $\mathcal{S}$               | T3 – B ( $\alpha = 1$ ) |
|-----------------------------|-------------------------|
| $\lambda_{S_1}^{(1)} m_1^2$ | $6$                     |
| $g_2^2 m_1^2$               | $-\frac{9}{2}$          |
| $g_1^2 m_1^2$               | $-\frac{3}{2}$          |
| $\lambda_{12}^{(1)} m_2^2$  | $4$                     |

$\beta_{m_2^2}$ 

| $\mathcal{S}$               | T3 – B ( $\alpha = 1$ ) |
|-----------------------------|-------------------------|
| $\lambda_{S_2}^{(1)} m_2^2$ | 6                       |
| $g_2^2 m_2^2$               | $-\frac{9}{2}$          |
| $g_1^2 m_2^2$               | $-\frac{27}{2}$         |
| $\lambda_{12}^{(1)} m_1^2$  | 4                       |

 $\beta_{\lambda_{HS_1}^{(1)}}$ 

| $\mathcal{S}$                              | T3 – B ( $\alpha = 1$ ) |
|--------------------------------------------|-------------------------|
| $ \lambda_5 ^2$                            | 6                       |
| $g_2^2 \lambda_{HS_1}^{(1)}$               | -9                      |
| $g_1^2 \lambda_{HS_1}^{(1)}$               | -3                      |
| $\lambda_1 \lambda_{HS_1}^{(1)}$           | 6                       |
| $\left(\lambda_{HS_1}^{(1)}\right)^2$      | 4                       |
| $\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(1)}$ | 6                       |
| $\lambda_{12}^{(1)} \lambda_{HS_2}^{(1)}$  | 4                       |
| $g_2^4$                                    | $\frac{9}{4}$           |
| $g_1^4$                                    | $\frac{3}{4}$           |
| $\left(\lambda_{HS_1}^{(A)}\right)^2$      | $\frac{3}{4}$           |
| $\lambda_{HS_1}^{(1)} T$                   | 2                       |

 $\beta_{\lambda_{HS_2}^{(1)}}$ 

| $\mathcal{S}$                              | T3 – B ( $\alpha = 1$ ) |
|--------------------------------------------|-------------------------|
| $ \lambda_5 ^2$                            | 6                       |
| $g_2^2 \lambda_{HS_2}^{(1)}$               | -9                      |
| $g_1^2 \lambda_{HS_2}^{(1)}$               | -15                     |
| $\lambda_1 \lambda_{HS_2}^{(1)}$           | 6                       |
| $\left(\lambda_{HS_2}^{(1)}\right)^2$      | 4                       |
| $\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(1)}$ | 6                       |
| $\lambda_{12}^{(1)} \lambda_{HS_1}^{(1)}$  | 4                       |
| $g_2^4$                                    | $\frac{9}{4}$           |
| $g_1^4$                                    | $\frac{27}{4}$          |
| $\left(\lambda_{HS_2}^{(A)}\right)^2$      | $\frac{3}{4}$           |
| $\lambda_{HS_2}^{(1)} T$                   | 2                       |

 $\beta_{\lambda_{12}^{(1)}}$ 

| $\mathcal{S}$                            | T3 – B ( $\alpha = 1$ ) |
|------------------------------------------|-------------------------|
| $ \lambda_5 ^2$                          | 3                       |
| $g_2^2 \lambda_{12}^{(1)}$               | -9                      |
| $g_1^2 \lambda_{12}^{(1)}$               | -15                     |
| $\left(\lambda_{12}^{(1)}\right)^2$      | 4                       |
| $\lambda_{12}^{(1)} \lambda_{S_1}^{(1)}$ | 6                       |

|                                             |                |
|---------------------------------------------|----------------|
| $\lambda_{12}^{(1)} \lambda_{S_2}^{(1)}$    | 6              |
| $\lambda_{HS_1}^{(1)} \lambda_{HS_2}^{(1)}$ | 4              |
| $g_2^4$                                     | $\frac{9}{4}$  |
| $g_1^4$                                     | $\frac{27}{4}$ |
| $\left(\lambda_{12}^{(A)}\right)^2$         | $\frac{3}{4}$  |

$\beta_{\lambda_5}$

| $S$                              | T3 – B ( $\alpha = 1$ ) |
|----------------------------------|-------------------------|
| $\lambda_5 \lambda_1$            | 2                       |
| $\lambda_5 \lambda_{HS_1}^{(1)}$ | 4                       |
| $\lambda_5 \lambda_{HS_2}^{(1)}$ | 4                       |
| $\lambda_5 \lambda_{12}^{(1)}$   | 2                       |
| $g_2^2 \lambda_5$                | −9                      |
| $g_1^2 \lambda_5$                | −9                      |
| $\lambda_5 \lambda_{HS_1}^{(A)}$ | −2                      |
| $\lambda_5 \lambda_{HS_2}^{(A)}$ | 2                       |
| $\lambda_5 \lambda_{12}^{(A)}$   | − $\frac{1}{2}$         |
| $\lambda_5 T$                    | 2                       |

## Additional scalar-coupling beta functions

$\beta_{\lambda_{HS_1}^{(A)}}$

| $S$           | T3 – B ( $\alpha = 1$ ) |
|---------------|-------------------------|
| $g_1^2 g_2^2$ | 6                       |

$\beta_{\lambda_{HS_2}^{(A)}}$

| $S$           | T3 – B ( $\alpha = 1$ ) |
|---------------|-------------------------|
| $g_1^2 g_2^2$ | 18                      |

$\beta_{\lambda_{12}^{(A)}}$

| $S$           | T3 – B ( $\alpha = 1$ ) |
|---------------|-------------------------|
| $g_1^2 g_2^2$ | 18                      |

## Full Weinberg-coefficient RGE in EFT1

Writing the canonical mixed LLSS coefficient as  $C_{12}$ , the coupled one-loop Weinberg equation used in the  $F$ -first EFT is

$$16\pi^2 \frac{dC_5}{d\ln\mu} = (2\lambda_1 - 3g_2^2 + 2T)C_5 \quad (1)$$

$$- \frac{3}{2} [(Y_e^\dagger Y_e)^T C_5 + C_5 (Y_e^\dagger Y_e)] \quad (2)$$

$$+ R_W \lambda_5 C_{12}. \quad (3)$$

Note:  $\frac{7}{2}$  thing isnt there if we integrate out our fermion F so its not here

# Direct $C_{12} \rightarrow C_5$ recoupling factor

For each representation the source term is

$$16\pi^2 \beta_{C_5} \supset R_W \lambda_5 C_{12}.$$

| $\mathcal{S}$ | T3 – B ( $\alpha = 1$ ) |
|---------------|-------------------------|
| $R_W$         | $\frac{4\sqrt{3}}{3}$   |

.

T3 – B ( $\alpha = 1$ )

1 independent complex matched structure displayed.

| complex matched operator                      | matched prefactor                | $SU(2)_L$ invariant                         |
|-----------------------------------------------|----------------------------------|---------------------------------------------|
| $(L_i^T C L_j) S_1^\dagger S_2 + \text{h.c.}$ | $\frac{\overline{y_1 y_2}}{M_F}$ | $\mathcal{C}_{LS_1 F} \mathcal{C}_{LS_2 F}$ |

# Next threshold

Further matching depends on the next field or fields selected by the threshold plan. Once all T3 fields are removed, their representation dependence is carried by the matched  $C_5$  boundary condition and the subsequent SMEFT running is universal.